

Constructing a symmetric $W(35, 25)$ weighing matrix and further H-classes

Christopher Munro
lattice.living535@passfwd.com

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github.com/Jepfore-Games-Ltd/W-35-25

Abstract

The existence of any weighing matrix $W(35, 25)$ has been listed as open in recent catalogues. I describe a block matrix construction with an order-five symmetry, encode the problem as a CP-SAT instance, and solve it with OR-Tools in Python. The solver produces a $W(35, 25)$ matrix, thereby confirming existence for these parameters. Moreover, it is symmetric and regular. I then complete the known block-circulant cores by allowing the two borders to vary independently, obtaining 472 H-classes, and free one further band from the circulant assumption to obtain 94 more. The resulting collection contains 566 confirmed H-classes of ranks over $\mathbb{F}_5$ from 8 through 17. The cases of ranks 6 and 7 remain questions for further study.

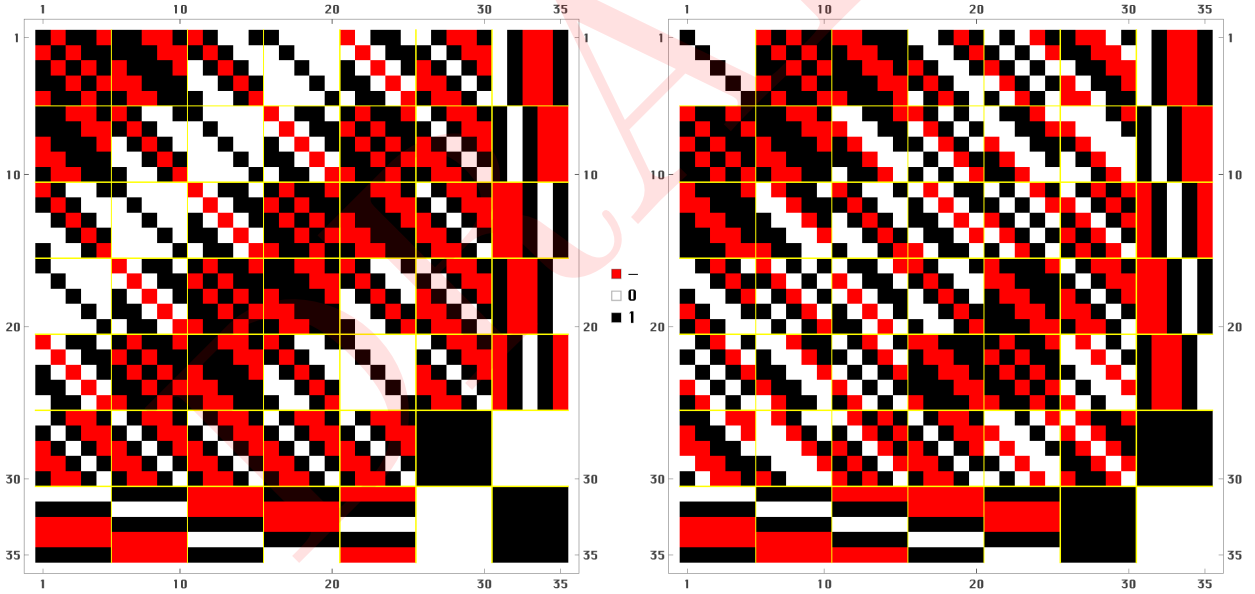


Figure 1: Symmetric weighing matrices $W(35, 25)$.
 $\Lambda = J$ (left), $\Lambda = \mathbf{0}$ (right)

1 Introduction

Weighing matrices are a class of combinatorial designs that generalise Hadamard matrices. They have been studied extensively in combinatorics, design theory, and related areas. A square *weighing matrix* W of order n and weight k , sometimes denoted by abuse of notation as

$W(n, k)$, is an $n \times n$ matrix with *ternary* entries w_{ij} in $\{0, 1, -1\}$ satisfying the *weighing equation*

$$WW^T = kI_n, \quad (1)$$

where I_n denotes the $n \times n$ identity matrix. This constrains every row of W to have exactly k non-zero entries. This paper focuses on finding a $W(35, 25)$ instance, which consists of 35 orthog-

onal rows, each having 25 entries equal to 1 or -1 and 10 entries equal to zero.

Determining the existence of weighing matrices for a given choice of (n, k) is a classical problem in the theory of orthogonal designs [1, 2, 3] and can be challenging, despite their simple definition. Recent progress in construction methods and computational search continues to resolve existence questions. For example, Ben-Av et al. in [4] introduced *partitioned group matrices* and resolved the four remaining $k = 16$ cases, namely $W(23, 16)$, $W(25, 16)$, $W(27, 16)$, and $W(29, 16)$. Their approach also produced a solution to the previously open case $W(102, 97)$. $W(35, 25)$ was listed as unknown in [5]. Lampio and Ganzhinov list it as the smallest unresolved existence case for a general weighing matrix [7].

2 Early attempts

Simply searching for $W(35, 25)$ by brute force is unrealistic due to the high dimensionality of the problem. While I do not intend to elaborate too much on my initial optimization experiments, I

will briefly note that I made a number of attempts at simulated annealing. I worked with a CUDA-accelerated block annealer over a quasi-product $IW(7, 25) \otimes X_{5 \times 5}$ that combined the 44 TH-classes of $IW(7, 25)$ classified in [5] with circulant blocks X . The search attempted to find a W by choosing an X and a $Y \in IW(7, 25)$ and forming:

$$W = Y \otimes X$$

$$\mathcal{G}(W) = \left\| WW^\top - kI \right\|_1$$

$$W^* = \arg \min_W \{ (1 - \lambda) \mathcal{G}(W) + \lambda \text{Con}(W) \}.$$

Here $\mathcal{G}(W)$ gives the Gram defect: the sum of the absolute values of the off diagonal entries of the Gram matrix of W . On the other hand, $\text{Con}(W)$ counts the number of non-orthogonal pairs. These searches repeatedly settled at sub-optimal basins. In an attempt to escape these plateaus, I dynamically varied λ during optimization. I noticed that I could drive down the number of conflicts, while accepting a higher L_1 score and vice versa.

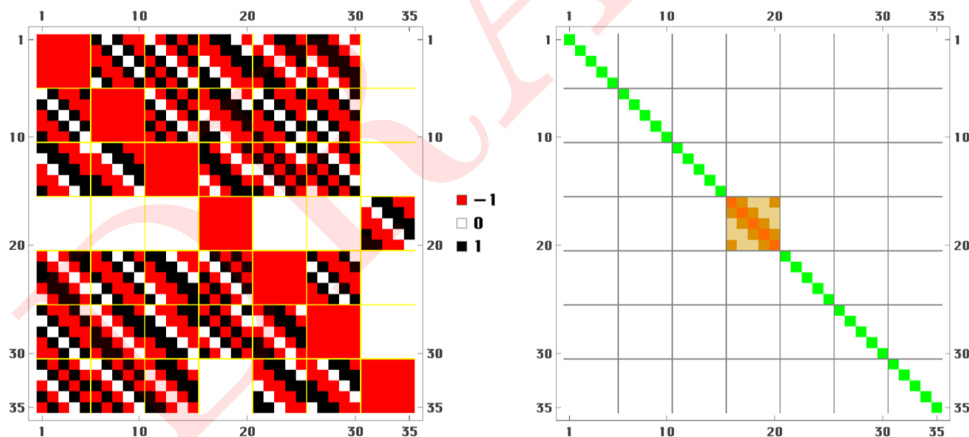


Figure 2: A near-miss result found through block annealing. The Gram matrix is almost a diagonal matrix, except at a single defective 5×5 block.

This led to a particularly close near-miss, though it did not satisfy the *weighing equation*. Further block annealing did not produce a valid weighing matrix, but visual inspection of the results inspired more tactical block arrangements. I considered whether additional constraints could make the problem more tractable for a CP-SAT solver, similar to constructions described in [4].

3 The construction

Goldberger et al. propose in [5] that solving open cases should begin by restricting to symmetric instances. The construction here is a bordered block-circulant matrix developed over $\mathbb{Z}_5$, which is a special case of the partitioned group matrices of Ben-Av et al in [4]. This construction is motivated by their work.

Start with an order-five symmetry σ . Label each of the first 30 rows and columns by

$$(\ell, a), \quad \ell = 0, \dots, 5, \quad a \in \mathbb{Z}_5.$$

Leave the final five rows and columns fixed. Then σ does this:

$$\sigma : (\ell, a) \mapsto (\ell, a + 1). \quad (2)$$

In other words, σ is the product of six disjoint 5-cycles and fixes indices 31, $\dots$, 35.

$$\underbrace{(1, 2, 3, 4, 5)}_S \underbrace{(6, 7, 8, 9, 10)}_S \cdots \underbrace{(26, 27, 28, 29, 30)}_S$$

where S is a permutation matrix

$$S = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix} \quad (3)$$

Index conventions: For reference throughout this paper, labels of blocks, of the columns of C and of the block rows and columns of Σ all start at 0, matching $\ell = 0, \dots, 5$ and $a \in \mathbb{Z}_5$. Actual row and column numbers of W start at 1 and run to 35, so that (ℓ, a) is row $5\ell + a + 1$ and the fixed rows are 31, $\dots$, 35.

Row and column permutations preserve the weighing-matrix equations, so consider a special case where P leaves W unchanged by simultaneous row-column permutation:

$$\begin{aligned} PWP^\top &= W \\ Q &= \text{diag}(S, S, S, S, S, S) \\ P &= \text{diag}(Q, I_5) \end{aligned} \quad (4)$$

Suppose a W could have this $5^6 1^5$ symmetry, and consider W as a block matrix:

$$W_{35 \times 35} = \begin{pmatrix} \Sigma_{30 \times 30} & \Psi_{30 \times 5} \\ \Psi_{5 \times 30}^\top & \Lambda_{5 \times 5} \end{pmatrix}. \quad (5)$$

Divide the first 30 columns into six groups of five. Then σ acts on Σ as $Q\Sigma Q^\top$. Entries within its 36 blocks are cyclically permuted. Also σ will permute $\Psi^\top$ to $\Psi^\top Q^\top$, and Λ is unaffected. Consequently, there are six $\psi_{5 \times 5}^\top$ blocks that are row-constant. Take one column c_i from each of the $\psi_i^\top$ and put them in a 5×6 matrix C .

For rows fixed by σ , each column of C is repeated five times. The dot products of those rows must satisfy the following equation, noting that the fixed rows are orthogonal with squared norm 25:

$$5CC^\top + \Lambda\Lambda^\top = 25I_5 \quad (6)$$

The rows in the concatenation $(\Psi^\top | \Lambda)$ must have 25 non-zero entries. An entry C_{ij} contributes 5 copies of itself to its respective row in the block $\psi_j^\top$. Define $\text{wt}(X_i)$ to be the number of non-zero entries in row X_i of a matrix X . Then

$$\begin{aligned} \text{wt}(\Psi_{r,*}^\top) + \text{wt}(\Lambda_{r,*}) &= 25 \\ 5 \text{wt}(C_{r,*}) + \text{wt}(\Lambda_{r,*}) &= 25. \end{aligned} \quad (7)$$

Each row of Λ must have weight 0 or 5. The same argument applies to columns: since W is square, $WW^\top = 25I$ also implies $W^\top W = 25I$. Then either Λ is the zero matrix or it has no zero entries.

Now for Λ try guessing

$$\Lambda = J_{5 \times 5}.$$

where $J_{5 \times 5}$ is a five-by-five matrix of 1s. Apply equation (6) and rearrange to get

$$CC^\top = 5I_5 - J_{5 \times 5} \quad (8)$$

The diagonal entries on the right are 4, while the off-diagonal entries are -1 . This forces each row of C to have Euclidean norm 2. Two different rows must have dot product -1 . There happens to be a way to construct such a C starting with

$$(0, 1, -1, -1, 1, 0).$$

Cyclically shift its first five entries to the right four times, always leaving the final zero fixed. Stacking the five resulting rows gives

$$C = \begin{pmatrix} 0 & 1 & -1 & -1 & 1 & 0 \\ 1 & 0 & 1 & -1 & -1 & 0 \\ -1 & 1 & 0 & 1 & -1 & 0 \\ -1 & -1 & 1 & 0 & 1 & 0 \\ 1 & -1 & -1 & 1 & 0 & 0 \end{pmatrix} \quad (9)$$

Keen observers will recognize this as a *Jacobsthal* matrix $\mathcal{J}_5$ with an extra zero column, using Paley's quadratic-character [11]. The *quadratic character* χ_5 generates entries in the first row:

$$\chi_5(a) = \left(\frac{a}{5}\right) \quad (10)$$

for a in $\{0, \dots, 4\}$, where the parenthesized fraction is the *Legendre symbol*.

The rows of C have four non-zero entries, and direct dot products give -1 between different rows.

Step 1: Build Ψ from C

Let c_i be column i of C , where $i = 0, \dots, 5$, and let $\mathbf{1}_5$ denote a column of five 1's. Each block $\psi_i^\top$ is given by $c_i \mathbf{1}_5^\top$.

$$\Psi^\top = (c_0 \mathbf{1}_5^\top \quad c_1 \mathbf{1}_5^\top \quad \dots \quad c_5 \mathbf{1}_5^\top). \quad (11)$$

Step 2: Build Σ from cyclic blocks

For a row of five numbers $a_0, \dots, a_4$, define $\text{Cyc}(a_0, a_1, a_2, a_3, a_4)$ as the matrix:

$$\begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ a_4 & a_0 & a_1 & a_2 & a_3 \\ a_3 & a_4 & a_0 & a_1 & a_2 \\ a_2 & a_3 & a_4 & a_0 & a_1 \\ a_1 & a_2 & a_3 & a_4 & a_0 \end{pmatrix} \quad (12)$$

Each row is a cyclic shift to the right of the row above, so Cyc produces a *circulant* block from a *seed* row. Let s_{ij} be the 5-tuple seed in position (i, j) of the table below. Define the (i, j) block of Σ to be $\text{Cyc}(s_{ij})$. The table gives a solution to all 36 of the 5×5 circulant block seeds of $\Sigma_{\Lambda=J}$:

	0	1	2	3	4	5
0	(1, −, 1, 1, −)	(1, 1, 1, −, −)	(0, 1, 1, 0, −)	(0, 0, 1, −, 1)	(0, 0, 1, 0, 0)	(0, 1, −, −, 1)
1	(1, −, −, 1, 1)	(−, 1, 0, 0, 1)	(1, 1, −, 1, −)	(0, 0, 1, 0, 0)	(0, −, 0, 1, 1)	(1, −, −, 1, 0)
2	(0, −, 0, 1, 1)	(1, −, 1, −, 1)	(1, 0, 0, 0, 0)	(1, −, −, 1, 1)	(0, 0, 1, −, 1)	(1, 0, 1, −, −)
3	(0, 1, −, 1, 0)	(0, 0, 0, 1, 0)	(1, 1, 1, −, −)	(−, 0, 1, 1, 0)	(−, 1, 1, −, 1)	(−, 1, 0, 1, −)
4	(0, 0, 0, 1, 0)	(0, 1, 1, 0, −)	(0, 1, −, 1, 0)	(−, 1, −, 1, 1)	(1, 1, −, −, 1)	(−, −, 1, 0, 1)
5	(0, 1, −, −, 1)	(1, 0, 1, −, −)	(1, −, −, 1, 0)	(−, −, 1, 0, 1)	(−, 1, 0, 1, −)	(1, 1, 1, 1, 1)

Table 1: The 36 Σ block “seeds” s_{ij} . Here, ‘−’ denotes -1 .

The table of block seeds was computed with a CP-SAT solver by imposing the usual weighing matrix constraints (see Appendix F). But note that since W is symmetric, it sufficed to find just 21 blocks in the upper triangle, with the extra constraint that the 6 blocks along the main diagonal must themselves be symmetric. Also $s_{ji} = (a_0, a_4, a_3, a_2, a_1)$ when $s_{ij} = (a_0, a_1, a_2, a_3, a_4)$. These extra constraints helped to cut down solution time.

4 The result

The constructed 35×35 matrix is a symmetric weighing matrix $W(35, 25)$. Simultaneously shifting the five indices inside each of the six block rows and the corresponding block columns gives an automorphism of order 5 that fixes the final five rows and columns. To verify the result, check that $WW^\top = 25I$, that all rows contain 25 entries of ± 1 and 10 zeros, and that since W is symmetric, the same holds for its columns.

Step 3: Assemble W

Finally, insert the completed matrices $\Sigma, \Psi, \Psi^\top$ into (5) with $\Lambda = J_{5 \times 5}$ and observe

$$\begin{aligned} \Sigma \Sigma^\top + \Psi \Psi^\top &= 25I_{30 \times 30} \\ \Sigma \Psi + \Psi \Lambda &= 0_{30 \times 5} \\ W &= W^\top \\ WW^\top &= 25I_{35 \times 35} \\ \text{wt}(W_r) &= 25, \quad r = 1, \dots, 35 \end{aligned} \quad (13)$$

Theorem 1. *A symmetric 35×35 weighing matrix with weight 25 exists.*

Proof. The matrix W displayed in Appendix A is a weighing matrix $W(35, 25)$. Substitute W into (1) and verify by direct computation. $\square$

One can also choose $\Lambda = \mathbf{0}$ and obtain another matrix in Appendix B. Constructing this matrix requires a different matrix C (*contrast with*

equation (9)) to satisfy equation (6):

$$C_{\Lambda=\mathbf{0}} = \begin{pmatrix} 1 & 0 & 1 & -1 & -1 & 1 \\ 1 & 1 & 0 & 1 & -1 & -1 \\ 1 & -1 & 1 & 0 & 1 & -1 \\ 1 & -1 & -1 & 1 & 0 & 1 \\ 1 & 1 & -1 & -1 & 1 & 0 \end{pmatrix}$$

Both constructions have blocks $\beta_{ji} = \beta_{ij}^\top$.

Observe that every row sum of $W_{\Lambda=\mathbf{0}}$ is 5, and likewise for $W_{\Lambda=J}$. Both are symmetric, so every column sum is 5 too.

A matrix with constant row and column sums is called *regular*. Additionally, since both matrices are symmetric, it follows that

$$W^2 = 25I.$$

5 Observations

Every eigenvalue of W is ± 5 . Regularity gives $W\mathbf{1} = 5\mathbf{1}$ so $\mathbf{1}$ is a $+5$ eigenvector. Direct counting yields $\text{tr}(W_{\Lambda=J}) = 15$ and $\text{tr}(W_{\Lambda=\mathbf{0}}) = 5$, and hence the spectra

$$(+5)^{19}(-5)^{16} \quad \text{and} \quad (+5)^{18}(-5)^{17}.$$

Each row (and each column) of either matrix has fifteen entries equal to $+1$, ten equal to -1 , and ten equal to 0 .

The diagonals are also interesting: $W_{\Lambda=J}$ has no zeros on the diagonal (25 entries $+1$ and 10 entries -1), while $W_{\Lambda=\mathbf{0}}$ has ten zeros on the diagonal.

I had hoped that a 7×7 construction with a similar block layout might be possible with 4 moving 7×7 blocks and a 7-wide constant border. However, with $\Lambda = J$, each fixed row already has 7 non-zeros and would need another 18 from $\Psi^\top$. Likewise, if $\Lambda = \mathbf{0}$ then each fixed row must get 25 non-zeros from $\Psi^\top$. Unfortunately, 7 divides neither 18 nor 25, rendering this particular 7×7 construction method impossible.

This rules out an order-seven symmetry at $n = 35$ and $k = 25$ only. It says nothing about the $p = 7$ member of the Jacobsthal family raised in the Questions section, which is a different object with $n = 63 = 7(7 + 2)$ and $k = 49$.

6 Distinct classes

Plain permutations P, Q where $W' = PWQ$ preserve the weighing equation, and so does the negation $W' = -W$. However, non-trivial changes to W are more interesting. By performing simple block negations, one can obtain new weighing matrices from the two CP-SAT solutions established so far. I use the notion of *Hadamard equivalence* as discussed in [4].

Given weighing matrices A, B and signed monomial permutation matrices P, Q , write:

$$A \sim_{\text{H}} B \iff \exists P, Q : PAQ^\top = B$$

where $\sim_{\text{H}}$ is an equivalence relation. If $A \sim_{\text{H}} B$, then A, B are said to be *Hadamard-equivalent*, abbreviated to *H-equivalent*. This $\sim_{\text{H}}$ partitions the set of all $W(n, k)$ into equivalence classes known as *H-classes*. Following [5, Definition 1.1], define the TH-equivalence relation by

$$A \sim_{\text{TH}} B \iff A \sim_{\text{H}} B \text{ or } A^\top \sim_{\text{H}} B.$$

Its equivalence classes are called *TH-classes*.

Let $\text{rank}_5(W)$ be the matrix rank of W over finite field $\mathbb{F}_5$. This number is invariant under signed row and column permutations. If A, B are weighing matrices, then

$$A \sim_{\text{H}} B \implies \text{rank}_5(A) = \text{rank}_5(B). \quad (14)$$

The contrapositive statement, different ranks imply inequivalence, gives an efficient way to detect $A \not\sim_{\text{H}} B$. Equal rank is not sufficient to determine equivalence:

$$\text{rank}_5(A) = \text{rank}_5(B) \not\Rightarrow A \sim_{\text{H}} B. \quad (15)$$

Why choose $\mathbb{F}_5$ and not some other field? After all, any field of characteristic different from 2, such as $\mathbb{F}_3$, also maintains rank under signed permutations. Unfortunately, under $\mathbb{F}_3$ the $W(35, 25)$ weighing equation (1) becomes:

$$WW^\top = I_{35} \quad \text{over } \mathbb{F}_3.$$

Consequently W is always invertible over $\mathbb{F}_3$ and hence $\text{rank}_3(W) = 35$. Being full rank means it cannot distinguish between matrices of different H-classes.

On the other hand, $\mathbb{F}_5$ is particularly nice to work with since

$$\begin{aligned} WW^\top &= 0 \quad \text{over } \mathbb{F}_5 \\ \text{Row}(W) &\subseteq \text{Row}(W)^\perp. \end{aligned}$$

That is, the row space $\mathcal{R} = \text{Row}(W)$ is totally self-orthogonal, contained in its orthogonal complement. Since the ambient space is $\mathbb{F}_5^{35}$,

$$\begin{aligned} \dim \mathcal{R} + \dim \mathcal{R}^\perp &= 35, \\ \text{rank}_5(W) = \dim \mathcal{R} &\leq \left\lfloor \frac{35}{2} \right\rfloor = 17. \end{aligned}$$

A second invariant is needed, because rank alone does not separate everything. One might reach for the Smith normal form, but for a $W(35, 25)$ it's not any better than using rank_5 .

Equation (1) gives $W^\top = 25W^{-1}$, so $25W^{-1}$ has integer entries and every elementary divisor d_i of W divides 25. Hence $d_i \in \{1, 5, 25\}$. Furthermore W and $W^\top$ have the same Smith form, and the Smith form of $25W^{-1}$ consists of the values $25/d_i$, so the multiset $\{d_i\}$ is closed under $d \mapsto 25/d$. That pairs the 1s up with the 25s in equal numbers. Since the elementary divisors coprime to 5 are exactly the 1s, their count is $\text{rank}_5(W)$, and writing $r = \text{rank}_5(W)$ leaves

$$\text{SNF}(W) = 1^r : 5^{35-2r} : 25^r. \quad (16)$$

So the Smith form being just a function of the rank gives us no more information to distinguish classes.

What does work is the *4-profile* of Cooper, Milas, and Wallis [8] which will be later introduced in Theorem 2. The tables below record its zero component $q(W) = \pi_W(0)$ together with the next component $\pi_W(1)$, and the triple

$$(\text{rank}_5(W), \pi_W(0), \pi_W(1))$$

takes a distinct value on each of the 41 classes listed. Only two pairs of classes agree on $(\text{rank}_5, \pi_W(0))$ alone, namely $\langle 31 \rangle$ with $\langle 40 \rangle$ and $\langle 38 \rangle$ with $\langle 41 \rangle$, and $\pi_W(1)$ separates both pairs. Theorem 2 covers the whole distribution π_W , so this is sufficient for proof of pairwise inequivalence. But I have included a further way to check by using the program in Appendix G on the two close pairs as an additional confirmation method.

Now let $\mathbf{W}_{\langle 1 \rangle}$ be the class containing representative $W_{\Lambda=J}$, and let $\mathbf{W}_{\langle 2 \rangle}$ be the class containing representative $W_{\Lambda=0}$. Negating particular blocks obtains new matrices that are representatives from distinct H-classes, and this gives Table 2: 8 H-classes across 6 distinct ranks.

The border-permutation and block-negation family described after Theorem 2 consists of exactly 40 H-classes and 34 TH-classes across 9 ranks. A separate CP-SAT class below gives a collection of at least 41 H-classes and at least 35 TH-classes, again across those 9 ranks. This is a first step towards a classification and is certainly not exhaustive. There may be many more classes.

W	Base	Blocks negated	$\text{rank}_5(W)$	$\pi_W(0)$	$\pi_W(1)$
$\langle 1 \rangle$	$W_{\Lambda=J}$	None.	10	5950	13910
$\langle 2 \rangle$	$W_{\Lambda=0}$	None.	17	6965	12440
$\langle 3 \rangle$	$W_{\Lambda=J}$	Λ .	9	5940	13800
$\langle 4 \rangle$	$W_{\Lambda=J}$	The unique J -block of Σ .	11	6050	13110
$\langle 5 \rangle$	$W_{\Lambda=J}$	Both of the above.	10	6040	13000
$\langle 6 \rangle^*$	$W_{\Lambda=0}$	The J -block of Ψ .	16	6910	12350
$\langle 7 \rangle^*$	$W_{\Lambda=0}$	The J -block of $\Psi^\top$.	16	6925	12510
$\langle 8 \rangle$	$W_{\Lambda=0}$	Both of the above.	15	6870	12420

Table 2: Distinct **W** H-classes. The base matrices are $W_{\Lambda=J}$ from Appendix A and $W_{\Lambda=0}$ from Appendix B.

The starred ($\star$) classes $\mathbf{W}_{\langle 6 \rangle}$ and $\mathbf{W}_{\langle 7 \rangle}$ are especially interesting. Matrices obtained from the recipes for these classes are not symmetric, but $\mathbf{W}_{\langle 6 \rangle} = \mathbf{W}_{\langle 7 \rangle}^\top$. Their rank_5 alone does not distinguish their classes. But their distinct zero components of the 4-profile, calculated be-

low, show that they are in different H-classes. Since their representatives are transposes of each other, these matrices belong to a TH-class, and this class contains no symmetric representative. I prove these claims in Theorem 3 below, using the zero component of the 4-profile from [8].

Theorem 2. Let $A = (a_{ij})$ be an $n \times n$ matrix and $[n] = \{1, \dots, n\}$. For a four-element subset $\{i, j, k, \ell\} \subseteq [n]$, write

$$\Pi_A(i, j, k, \ell) = \sum_{x=1}^n a_{ix}a_{jx}a_{kx}a_{\ell x}.$$

In the terminology of [8], let π_A denote the 4-profile distribution of these four-row sums, adapted here to ternary matrices by recording their absolute values:

$$\pi_A(v) := \# \left\{ \{i, j, k, \ell\} \in \binom{[n]}{4} : |\Pi_A(i, j, k, \ell)| = v \right\}$$

for $v = 0, 1, \dots, n$. Write $q(A) := \pi_A(0)$ for its zero component. Then the entire distribution π_A , and in particular $q(A)$, is invariant under Hadamard equivalence $\sim_H$.

Proof. Permuting rows permutes the four-element subsets, while permuting columns reorders the terms in each sum. Suppose m of the four selected rows are negated, where $0 \leq m \leq 4$. Every term of $\Pi_A(i, j, k, \ell)$ then acquires the same factor $(-1)^m$, so the sum is multiplied by $(-1)^m$ and its absolute value is unchanged. Negating any of the other $n - 4$ rows leaves this sum untouched, since those rows do not appear in it. A column negation has no effect because its sign occurs four times and contributes $(-1)^4 = 1$. Signed permutations therefore match up the four-element subsets of A and of $PAQ^\top$ in a way that preserves $|\Pi|$, so $\pi_{PAQ^\top} = \pi_A$, and in particular $q(PAQ^\top) = q(A)$, for signed permutation matrices P, Q . $\square$

$\pi_{A^\top}$ is also H-invariant. Transposing $PAQ^\top = B$ gives $QA^\top P^\top = B^\top$, so transposition preserves H-equivalence and Theorem 2 applies to the transposes.

Theorem 3. $\mathbf{W}_{\langle 6 \rangle} \cap \mathbf{W}_{\langle 7 \rangle} = \emptyset$ and if $M = M^\top$ then $M \notin \mathbf{W}_{\langle 6 \rangle} \cup \mathbf{W}_{\langle 7 \rangle}$.

Proof. First, we have established representatives $U \in \mathbf{W}_{\langle 6 \rangle}$ and $V \in \mathbf{W}_{\langle 7 \rangle}$ with $U = V^\top$ by following the recipe from the table. Direct calculation gives

$$q(U) = 6910, \quad q(V) = q(U^\top) = 6925.$$

Since $q(U) \neq q(V)$, Theorem 2 gives $U \not\sim_H V$, and hence

$$\mathbf{W}_{\langle 6 \rangle} \cap \mathbf{W}_{\langle 7 \rangle} = \emptyset.$$

Now suppose U has a symmetric matrix M in its H-class. Then

$$U \sim_H M = M^\top \sim_H U^\top = V \implies U \sim_H V$$

which is a contradiction, so $\mathbf{W}_{\langle 6 \rangle}$ has no symmetric representative. Interchanging U and V in that argument rules one out in $\mathbf{W}_{\langle 7 \rangle}$ as well. Thus the two H-classes are distinct, and neither contains a symmetric representative. $\square$

The two components $\pi_W(0)$ and $\pi_W(1)$, together with rank_5 , separate the further classes below.

Let $c_0, \dots, c_5$ be the columns of the corresponding matrix C , and let $e = 5$ when $\Lambda = J$, while $e = 0$ when $\Lambda = \mathbf{0}$. Write Ψ_{id} for the border whose block row i is $\mathbf{1}_5 c_i^\top$, with $c_e = \mathbf{0}$ in the first case and $c_e = \mathbf{1}_5$ in the second. For a permutation τ of $\{0, \dots, 5\}$ fixing e , define

$$\Psi_\tau = \begin{pmatrix} \mathbf{1}_5 c_{\tau(0)}^\top \\ \vdots \\ \mathbf{1}_5 c_{\tau(5)}^\top \end{pmatrix}, \quad W_\tau = \begin{pmatrix} \Sigma & \Psi_\tau \\ \Psi_\tau^\top & \Lambda \end{pmatrix}.$$

The 960-matrix family uses this one common τ in the two border positions. Their orders are not chosen independently, and Σ remains fixed.

Which Σ stays fixed matters, so let me pin the two halves of the family down. The $\Lambda = J$ half uses C from (9) and takes Σ to be the top-left 30×30 submatrix of the neater representative in Appendix E. The $\Lambda = \mathbf{0}$ half uses $C_{\Lambda=\mathbf{0}}$ and takes Σ to be the top-left 30×30 submatrix of Appendix B. In each half τ runs over the $5! = 120$ permutations of $\{0, \dots, 5\}$ that fix e , so nothing but the border order and the block signs changes.

The four block negations are not the same on the two sides, since the two borders put their all-ones block in different places. With $\Lambda = J$ the matrix C of (9) has a zero column and five Jacobsthal columns, so Ψ has no all-ones block at all. The two negatable all-ones blocks are Λ itself and the single J -block of Σ , which gives the four choices: nothing, Λ , that Σ block, and both. With $\Lambda = \mathbf{0}$, negating Λ does nothing, but $C_{\Lambda=\mathbf{0}}$ does have an all-ones column, so Ψ has a J -block. The four choices there are: nothing, that block of Ψ , the matching block of $\Psi^\top$, and both. All four preserve (1) on either side, though the middle two on the $\Lambda = \mathbf{0}$ side destroy symmetry.

For $i, j \neq e$, the Jacobsthal columns satisfy $c_i^\top c_i = 4$ and $c_i^\top c_j = -1$ when $i \neq j$, while c_e is orthogonal to them. Hence

$$\Psi_\tau \Psi_\tau^\top = \Psi_{\text{id}} \Psi_{\text{id}}^\top, \quad \Psi_\tau^\top \Psi_\tau = \Psi_{\text{id}}^\top \Psi_{\text{id}}. \quad (17)$$

Let $R = (\rho_{ij})$ record the common row sum of each 5×5 block Σ_{ij} . For the two cores,

$$\begin{aligned} R_{\Lambda=J} &= \text{diag}(J_{5 \times 5}, 5), \\ R_{\Lambda=\mathbf{0}} &= \text{diag}(0, J_{5 \times 5}). \end{aligned} \quad (18)$$

Since $\sum_{j \neq e} c_j = \mathbf{0}$ and every exchangeable c_j has coordinate sum zero, these forms of R give, in block row i ,

$$(\Sigma \Psi_\tau + \Psi_\tau \Lambda)_i = \mathbf{1}_5 \left(\sum_{j=0}^5 \rho_{ij} c_{\tau(j)}^\top + c_{\tau(i)}^\top \Lambda \right) = \mathbf{0}. \quad (19)$$

Together with (6) and the top-left block equation for the base matrix, these identities prove

$$\begin{aligned} W_\tau W_\tau^\top &= 25 I_{35}, \\ W_\tau &= W_\tau^\top, \\ W_\tau \mathbf{1}_{35} &= 5 \mathbf{1}_{35}. \end{aligned} \quad (20)$$

Thus every allowed common border permutation preserves both the weighing equation and regularity.

6.1 Ψ -block permutation of $\langle 1 \rangle$

On the neater representative of $\langle 1 \rangle$ (Appendix E), one of the six blocks is zero. Keep that zero block where it is, and permute the other five. Write $W_{\langle 1 \rangle}(\pi)$ for the matrix obtained by stacking those five blocks of Ψ in the order π . The neater matrix in the appendix uses $\pi = (0, 1, 3, 4, 2)$, which stays in $\langle 1 \rangle$. So the two representatives are in $\langle 1 \rangle$ here: $W_{\langle 1 \rangle}(0, 1, 3, 4, 2)$ and the matrix of Appendix A. They are different matrices, since their cores Σ differ, but they share $(\text{rank}_5, \pi_W(0), \pi_W(1)) = (10, 5950, 13910)$, and a signed-equivalence check with Appendix G can find the P, Q that relates them. So both are representatives of the one class. However, these three orders do not stay in the $\langle 1 \rangle$ class

$$\begin{aligned} \pi_{\text{id}} &= (0, 1, 2, 3, 4), \\ \pi_t &= (0, 1, 3, 2, 4), \\ \pi_c &= (0, 1, 2, 4, 3). \end{aligned} \quad (21)$$

The Λ and J -block negations of Table 2 applied to these three matrices give Table 3, including a symmetric representative of rank_5 8.

6.2 Ψ -block permutation of $\langle 2 \rangle$

On $\langle 2 \rangle$, one of the six blocks is J . Keep that J -block in place, and permute the other five. Write $W_{\langle 2 \rangle}(\tau)$ for the matrix whose six Ψ -blocks appear in the order τ , with $\tau_0 = 0$. The same J -block negations as $\langle 6 \rangle$ – $\langle 8 \rangle$ then give Table 4.

W	Base	Blocks negated	$\text{rank}_5(W)$	$\pi_W(0)$	$\pi_W(1)$
$\langle 9 \rangle$	π_{id}	Λ and the unique J -block of Σ .	8	6850	12650
$\langle 10 \rangle$	π_{id}	Λ .	9	6750	13450
$\langle 11 \rangle$	π_{id}	The unique J -block of Σ .	9	7000	12300
$\langle 12 \rangle$	π_{c}	None.	10	5940	13770
$\langle 13 \rangle$	π_{t}	None.	10	5955	13895
$\langle 14 \rangle$	π_{t}	Λ and the unique J -block of Σ .	10	6035	13015
$\langle 15 \rangle$	π_{c}	Λ and the unique J -block of Σ .	10	6050	13140
$\langle 16 \rangle$	π_{id}	None.	10	6900	13100
$\langle 17 \rangle$	π_{t}	Λ .	11	5935	13815
$\langle 18 \rangle$	π_{c}	Λ .	11	5950	13940
$\langle 19 \rangle$	π_{c}	The unique J -block of Σ .	11	6040	12970
$\langle 20 \rangle$	π_{t}	The unique J -block of Σ .	11	6055	13095

Table 3: Further H-classes from Ψ -block permutation of the neater $\Lambda = J$ representative. Each base is the matrix $W_{\langle 1 \rangle}(\pi)$ of Appendix E for the border order named in the second column. The published order $\pi = (0, 1, 3, 4, 2)$ is not listed here: its four negations give $\langle 1 \rangle$, $\langle 3 \rangle$, $\langle 4 \rangle$, $\langle 5 \rangle$ of Table 2.

W	Base τ	Blocks negated	$\text{rank}_5(W)$	$\pi_W(0)$	$\pi_W(1)$
$\langle 21 \rangle$	$(0, 1, 3, 4, 2, 5)$	None.	13	6975	12615
$\langle 22 \rangle^*$	$(0, 1, 3, 4, 2, 5)$	The J -block of Ψ .	14	6900	12675
$\langle 23 \rangle^*$	$(0, 1, 3, 4, 2, 5)$	The J -block of $\Psi^\top$.	14	7045	12565
$\langle 24 \rangle$	$(0, 1, 2, 3, 5, 4)$	None.	15	6830	12670
$\langle 25 \rangle$	$(0, 1, 3, 2, 4, 5)$	None.	15	6920	12360
$\langle 26 \rangle$	$(0, 1, 2, 5, 4, 3)$	None.	15	6970	12625
$\langle 27 \rangle$	$(0, 1, 3, 4, 5, 2)$	None.	15	6995	12505
$\langle 28 \rangle$	$(0, 1, 2, 4, 3, 5)$	None.	15	7045	12510
$\langle 29 \rangle^*$	$(0, 1, 3, 2, 4, 5)$	The J -block of Ψ .	16	6735	12400
$\langle 30 \rangle^*$	$(0, 1, 2, 4, 3, 5)$	The J -block of Ψ .	16	6820	12490
$\langle 31 \rangle^*$	$(0, 1, 2, 3, 5, 4)$	The J -block of Ψ .	16	6845	12650
$\langle 32 \rangle^*$	$(0, 1, 3, 2, 4, 5)$	The J -block of $\Psi^\top$.	16	6965	12385
$\langle 33 \rangle^*$	$(0, 1, 2, 4, 3, 5)$	The J -block of $\Psi^\top$.	16	6995	12625
$\langle 34 \rangle^*$	$(0, 1, 2, 3, 5, 4)$	The J -block of $\Psi^\top$.	16	7015	12735
$\langle 35 \rangle^*$	$(0, 1, 3, 4, 5, 2)$	The J -block of Ψ .	16	7260	12475
$\langle 36 \rangle$	$(0, 1, 3, 2, 5, 4)$	None.	17	6770	12605
$\langle 37 \rangle$	$(0, 1, 2, 4, 5, 3)$	None.	17	6780	12425
$\langle 38 \rangle$	$(0, 1, 3, 5, 4, 2)$	None.	17	7030	12715
$\langle 39 \rangle$	$(0, 1, 2, 5, 3, 4)$	None.	17	7110	12290
$\langle 40 \rangle^*$	$(0, 1, 3, 4, 5, 2)$	The J -block of $\Psi^\top$.	16	6845	12320

Table 4: Further H-classes from Ψ -block permutation of $\langle 2 \rangle$. Each base is $W_{\langle 2 \rangle}(\tau)$ for the border order τ in the second column. Starred representatives are nonsymmetric: $\langle 22 \rangle = \langle 23 \rangle^\top$, $\langle 29 \rangle = \langle 32 \rangle^\top$, $\langle 30 \rangle = \langle 33 \rangle^\top$, $\langle 31 \rangle = \langle 34 \rangle^\top$, and $\langle 35 \rangle = \langle 40 \rangle^\top$. The same J -block negations applied to $\langle 2 \rangle$ itself give $\langle 6 \rangle$ – $\langle 8 \rangle$.

Classes $\langle 31 \rangle$ and $\langle 40 \rangle$ are the closest pair in the table: they share $\text{rank}_5 = 16$ and $q = 6845$. The next component of the 4 -profile settles it, since

$$\pi_{W_{\langle 31 \rangle}}(1) = 12650,$$

$$\pi_{W_{\langle 40 \rangle}}(1) = 12320,$$

so Theorem 2 puts them in different H-classes. A complete signed-equivalence check using Appendix G also returned DISTINCT.

There are $5! = 120$ allowed border permutations for each choice of Λ , and four block-negation choices for each permutation, so the family holds $2 \cdot 120 \cdot 4 = 960$ matrices. I formed all 960 of them and, since that is small enough to be exhaustive, was able to count and check all of them. First, evaluating $(\text{rank}_5, \pi_W(0), \pi_W(1))$ on all 960 yields 40 distinct values, so by Theorem 2 there are at least 40 H-classes. Second, taking one representative per value and running the check of Appendix G on every remaining

member against the representative sharing its value, all 920 of those checks returned a signed-permutation pair, so there are at most 40. Split by border, this is exactly 16 H-classes and 16 TH-classes for $\Lambda = J$, and exactly 24 H-classes and 18 TH-classes for $\Lambda = \mathbf{0}$. Thus, this family contains exactly 40 H-classes and 34 TH-classes.

An extra $\Lambda = \mathbf{0}$ class

A special symmetric class $\langle 41 \rangle$ was found by CP-SAT, lying outside the 960-matrix family. Its representative is displayed in Appendix C. The following seeds generate it. Let c_j be column j of the displayed matrix $C_{\Lambda=\mathbf{0}}$, in the order $j = 0, \dots, 5$. Set $\Lambda = \mathbf{0}$, take block row j of Ψ to be $\mathbf{1}_5 c_j^\top$, and set the (i, j) block of Σ to $\text{Cyc}(s_{ij})$, where s_{ij} is the seed in row i , column j of Table 5. Assemble $W_{\langle 41 \rangle}$ using (5). In particular, the Ψ -block order is $(0, 1, 2, 3, 4, 5)$, with the all-one block first.

	0	1	2	3	4	5
0	(0, 1, −, −, 1)	(−, 1, 0, 0, 0)	(1, 0, −, 1, −)	(−, 1, 0, 1, −)	(0, −, −, 1, 1)	(1, 0, −, 0, 0)
1	(−, 0, 0, 0, 1)	(1, −, 1, 1, −)	(1, −, −, 1, 1)	(0, 1, 1, −, 0)	(1, 0, 1, −, 0)	(1, 1, 0, 0, −)
2	(1, −, 1, −, 0)	(1, 1, 1, −, −)	(1, 0, 0, 0, 0)	(1, 1, −, 1, −)	(1, 0, 0, −, 1)	(−, 0, 1, 1, 0)
3	(−, −, 1, 0, 1)	(0, 0, −, 1, 1)	(1, −, 1, −, 1)	(−, 1, 0, 0, 1)	(1, 0, −, 0, 1)	(0, −, 0, 1, 1)
4	(0, 1, 1, −, −)	(1, 0, −, 1, 0)	(1, 1, −, 0, 0)	(1, 1, 0, −, 0)	(−, 1, 0, 0, 1)	(1, −, 1, −, 1)
5	(1, 0, 0, −, 0)	(1, −, 0, 0, 1)	(−, 0, 1, 1, 0)	(0, 1, 1, 0, −)	(1, 1, −, 1, −)	(1, −, 1, 1, −)

Table 5: The 36 circulant seeds s_{ij} for $W_{\langle 41 \rangle}$, using the convention of Table 1.

$W_{\langle 41 \rangle}$ has these properties

$$\begin{aligned}
W_{\langle 41 \rangle} &= W_{\langle 41 \rangle}^\top, \\
W_{\langle 41 \rangle} W_{\langle 41 \rangle}^\top &= 25I, \\
\text{rank}_5(W_{\langle 41 \rangle}) &= 17, \\
q(W_{\langle 41 \rangle}) &= \pi_{W_{\langle 41 \rangle}}(0) = 7030, \\
\pi_{W_{\langle 41 \rangle}}(1) &= 12780.
\end{aligned} \tag{22}$$

Note the similarity to $\langle 38 \rangle$, which is the only one of $\langle 1 \rangle$ – $\langle 40 \rangle$ to share both $\text{rank}_5 = 17$ and $q = 7030$. But $\pi_{W_{\langle 38 \rangle}}(1) = 12715$, so Theorem 2 separates the two, and $\langle 41 \rangle$ already differs from each of the other 39 classes in $(\text{rank}_5, \pi_W(0))$. Hence $\langle 41 \rangle$ is H-inequivalent to all 40 classes in the family. Checking $\langle 41 \rangle$ against $\langle 38 \rangle$ for signed equivalence using Appendix G independently returned DISTINCT. Adding it gives a col-

lection of at least 41 H-classes and at least 35 TH-classes.

7 Beyond the first 41 classes

The question raised by $\langle 41 \rangle$ has a surprisingly large answer. It is not just one exceptional matrix with a few further block permutations. The weighing equation leaves more freedom in the border than the original construction uses, and it also leaves more freedom in one of the neighbouring bands. These two observations give 472 classes from the known block-circulant cores and a further 94 classes outside that family.

Completing a fixed core

The original construction is symmetric, so it places $\Psi^\top$ below Ψ . For this part it is useful not to impose symmetry. Write instead

$$W = \begin{pmatrix} \Sigma & \Psi \\ \Phi & \Lambda \end{pmatrix}, \quad \Psi \in \mathbb{Z}^{30 \times 5}, \quad \Phi \in \mathbb{Z}^{5 \times 30}. \quad (23)$$

For each of the symmetric cores used here, the two diagonal blocks of the weighing equation give

$$25I_{30} - \Sigma^2 = \Psi\Psi^\top = \Phi^\top\Phi. \quad (24)$$

Thus the core fixes the complete Gram matrix of both borders. It does not say that the two Gram factors must be transposes of one another.

For a $\Lambda = \mathbf{0}$ core, the remaining equations say $\Sigma\Psi = 0$ and $\Sigma\Phi^\top = 0$. The columns of Ψ and of $\Phi^\top$ have a remarkable property: they give orthogonal ternary bases of $\ker \Sigma$ (*its nullspace*), each vector having norm 25. There are exactly 12 such bases up to signed permutation of these five border blocks. The two bases can be cho-

sen independently, giving a 12×12 grid of 144 completions.

Core-completion recipe. Start with one of the known 30×30 cores Σ .

1. Compute $T = 25I_{30} - \Sigma^2$.
2. Find the ternary matrices $X \in \{0, \pm 1\}^{30 \times 5}$ satisfying $XX^\top = T$, keeping one representative under signed column permutation.
3. Choose two representatives X, Y independently and put $\Psi = X$, $\Phi = Y^\top$.
4. Insert the corresponding Λ and check the two linear cross equations from $WW^\top = W^\top W = 25I$.

For $\Lambda = \mathbf{0}$, every one of the 12^2 pairs completes the matrix.

The $\Lambda = J$ side is similar, except that the border has rank four and the corner is nonzero. It has 72 completions per core variant before H-equivalence is taken. Table 6 gives the complete count for the cores currently known.

Core	Λ	Completions	H-classes	Comment
I	$\mathbf{0}$	144	144	generic
II	$\mathbf{0}$	144	144	contains $\langle 41 \rangle$
III	J	72	12	$\Lambda = J$
III _n	J	72	12	negated core variant
VII	$\mathbf{0}$	144	16	extra equivalences
IX	$\mathbf{0}$	144	144	generic
Total		720	472	

Table 6: Completion counts for the known cores.

The 472 H-classes include all 41 classes above, so 431 of them are new. Published class $\langle 41 \rangle$ is representative H455, on core II. Applying only border permutations and all-one-block negations to that core gives 24 classes. Allowing the upper and lower borders to vary independently gives all 144 classes on that core. In this sense, the earlier family was a diagonal slice of the full completion table.

Most classes are separated by the three invariants

$$\text{rank}_5(W), \quad \pi_W(0), \quad \pi_W(1),$$

but their triple is not complete on the enlarged

collection. Every group sharing a triple was therefore checked directly for signed equivalence. No triple is shared between different cores discovered so far.

Doubling the constant border

One tempting extension is to replace the five-wide constant border by two such bands. This does not work. Suppose a block-circulant core has b block rows of size 5, and a block-constant border of width K , where $5b + K = 35$. Let $V \in \{0, \pm 1\}^{b \times K}$ record the value of the border on each block row. Taking block-row sums in the

upper-left part of the weighing equation gives

$$RR^\top + 5VV^\top = 25I_b.$$

Hence every row of V has at most five nonzero entries. For the doubled border, $K = 10$ and $b = 5$, so V has at most 25 nonzeros.

On the column side,

$$5V^\top V + \Lambda^\top \Lambda = 25I_{10}.$$

Each column of Λ has at most 10 nonzeros. The diagonal equation therefore forces every column of V to have at least three nonzeros, giving at least 30 in total. The contradictory bounds $30 \leq \text{wt}(V) \leq 25$ prove that two block-constant border bands are impossible.

Freeing the middle band

The failed doubled-border idea suggests a different reading of the same matrix. Split its rows and columns as $35 = 25 + 5 + 5$:

$$W = \left(\begin{array}{c|cc} S & A & B \\ \hline Y^\top & * & * \\ \hline Z^\top & * & * \end{array} \right). \quad (25)$$

where S is 25×25 and A, B, Y, Z are 25×5 . In the original construction, B and Z come from the constant border, while A and Y are assembled from circulant blocks. The latter condition is not required by the weighing equation.

Fix S, B, Z . The two Gram matrices of the middle bands are then forced:

$$\begin{aligned} AA^\top &= 25I_{25} - SS^\top - BB^\top, \\ YY^\top &= 25I_{25} - S^\top S - ZZ^\top. \end{aligned} \quad (26)$$

Each row of A or Y is only a ternary vector of length five. Signed permutations of the five columns give equivalent matrices and are discarded. Then all that remains is to find the 10×10 corner, marked by stars in (25).

This re-banding produced 94 further H-classes whose invariant triples do not occur among the 472. None has a 30×30 top-left block that is again a 6×6 array of circulants, so these matrices genuinely lie outside the family used in the first part of this paper. Their ranks are shown in Table 7.

rank ₅	Known-core completions	Re-banded	Combined
8	1	0	1
9	7	0	7
10	19	9	28
11	13	32	45
12	0	38	38
13	3	15	18
14	36	0	36
15	138	0	138
16	180	0	180
17	75	0	75
Total	472	94	566

Table 7: The ranks of the 566 confirmed H-classes in the combined collection.

The 38 matrices in the rank-12 row are the first rank-12 representatives in this collection, closing the only gap between 8 and 17. Appendix D displays one of them. The number 94 is conservative: matrices sharing the same invariant triple were represented only once, so no claim is made that re-banding produces *only* 94 classes. Altogether the two catalogues contain 566 confirmed H-classes, 525 of which are new relative to the 41 listed above.

The remaining low ranks

The combined catalogue realizes every rank from 8 through 17, but it contains no representative of rank 6 or 7. The constructions above do not decide whether either value can occur. I therefore leave both cases as open questions.

Data and verification

Listing 566 matrices in the paper would obscure the two short recipes above. The complete catalogue and the scripts printed in this paper are in the companion repository linked beneath the title. Representatives H001 through H566 are stored in the three directories `representatives/published/`, `representatives/completions/`, and `representatives/rebanded/`. The cores are stored in `cores/`. The file `representatives/manifest.csv` records each construction, its published and original identifiers, rank, 4-profile values, and symmetry.

The standalone script `scripts/verify_catalogue.py` re-reads every representative, checks both weighing identities, recomputes the rank and full row and column 4-profiles, and checks pairwise H-inequivalence. It separates most pairs by those exact invariants and applies an exact signed-incidence graph isomorphism test to every pair that remains. For the present catalogue, the invariants separate 159879 of the $\binom{566}{2} = 159895$ pairs. The exact graph test finds each of the remaining 16 pairs distinct.

8 Conclusion

A symmetric, regular $W(35, 25)$ exists, which is Theorem 1. Appendices A and B display two of them. Both came out of CP-SAT instances built on the bordered block-circulant construction over $\mathbb{Z}_5$, and together they settle the existence question for parameters listed as unknown in [5] and reported as the smallest unresolved case in [7]. Each has an automorphism of order five, constant row and column sums 5, and satisfies $W^2 = 25I$.

The original 40-class family and the extra class $\langle 41 \rangle$ are the starting point of a much larger collection. Completing the known cores with independent upper and lower borders gives exactly

472 H-classes within that construction. Re-banding then gives at least 94 further classes outside the block-circulant family, for 566 confirmed H-classes in total. The stored representatives realize every value of rank_5 from 8 through 17. In particular, re-banding supplies 38 representatives of the previously missing rank 12.

The combined catalogue contains no representatives of ranks 6 or 7, and their existence is not decided here. These counts are not asserted to classify all $W(35, 25)$: more cores or further non-circulant constructions may produce additional classes. The enlarged catalogue also shows why direct equivalence checking remains useful. The invariant triple separates most, but not all, of the 472 known-core classes.

9 Questions

These results immediately suggest some directions to explore:

- Does a $W(35, 25)$ of rank 6 or 7 over $\mathbb{F}_5$ exist? The known matrices realize every rank from 8 through 17, but neither of these two lower ranks occurs in the catalogues above.
- Are there further inequivalent symmetric $W(35, 25)$ matrices beyond the collection above, either among other CP-SAT outputs or after dropping the order-five symmetry?
- The same border is available with a Jacobsthal matrix of order p , for every odd prime p . One takes $p + 1$ orbits of length p together with p fixed rows, giving parameters $n = p(p + 2)$ and $k = p^2$, and the two natural choices $C = (\mathcal{J}_p \mid \mathbf{0})$ and $C = (\mathbf{1} \mid \mathcal{J}_p)$. The $p = 3$ case $W(15, 9)$ is already known and classified in [6]. No verified construction of $W(63, 49)$ was found in the literature. Its general status is unresolved, though no circulant $W(63, 49)$ exists. Leung and Ma rule out the *proper* circulant matrices here, since $7 \equiv 3 \pmod{4}$ divides 63 [9], and an improper one is impossible for the cheaper reason that its 49 non-zero entries would have to sit inside a subgroup of order at most 21. Could it fall to this construction method by setting $p = 7$?

- The 94 re-banded representatives are counted conservatively by their invariant triples. Can matrices sharing a triple be resolved into still more H-classes, or can other choices of the $25 + 5 + 5$ split produce further classes?
- A small remark: for an odd prime p , the Jacobsthal identities

$$\begin{aligned}\mathcal{J}_p \mathcal{J}_p^\top &= pI - J \\ \mathcal{J}_p \mathbf{1} &= \mathbf{0}\end{aligned}$$

make both choices of C satisfy the fixed-

row equation with $k = p^2$, independently of any further search. These identities establish only that fixed-row equation, not the existence of a full $W(p(p+2), p^2)$. The $(p+1) \times (p+1)$ array of $p \times p$ circulant core blocks still has to be filled so that the first $p(p+1)$ rows become pairwise orthogonal. For $p = 5$ that search is the CP-SAT instance of Appendix F. But it would be interesting to know for which primes it remains feasible, and whether the two borders again produce H-inequivalent solutions.

AI use declaration

The author declares that Large Language Models *Claude Opus 5* by *Anthropic* and *GPT-5.6 Sol* by *OpenAI* were used in the preparation of this work to:

- Explore the existing literature on weighing matrices.
- Review code and fix dimensionality mistakes in the paper.
- Frame the author's findings in terms of the symmetry $P = \text{diag}(Q, I_5)$.
- Obtain missing representatives in the Ψ -block permutation tables and help with the Ψ -block permutation enumeration argument.
- Generalize the border completion, formulate the re-banding construction, and audit the resulting catalogues.

A Choosing $\Lambda = J$

1	-	1	1	-	1	1	1	-	0	1	1	0	-	0	0	1	-	1	0	0	1	0	0	0	1	-	-	1	0	1	-	-	1			
-	1	-	1	1	-	1	1	1	-	0	1	1	0	1	0	0	0	1	-	0	0	0	1	0	1	0	1	-	-	0	1	-	-	1		
1	-	1	-	1	-	-	1	1	1	0	-	0	1	1	-	1	0	0	1	0	0	0	0	1	-	1	0	1	-	0	1	-	-	1		
1	1	-	1	-	1	-	-	1	1	1	0	-	0	1	1	-	1	0	0	1	0	0	0	0	-	-	1	0	1	0	1	-	-	1		
-	1	1	-	1	1	1	-	-	1	1	1	0	-	0	0	1	-	1	0	0	1	0	0	0	1	-	-	1	0	0	1	-	-	1		
1	-	-	1	1	-	1	0	0	1	1	1	-	1	-	0	0	1	0	0	0	-	0	1	1	1	-	-	1	0	1	0	1	-	-		
1	1	-	-	1	1	-	1	0	0	-	1	1	-	1	0	0	0	1	0	1	0	-	0	1	0	1	-	-	1	1	0	1	-	-		
1	1	1	-	-	0	1	-	1	0	1	-	1	1	-	0	0	0	0	1	1	1	0	-	0	1	0	1	-	-	1	0	1	-	-		
-	1	1	1	-	0	0	1	-	1	-	1	-	1	1	1	0	0	0	0	0	1	1	0	-	-	1	0	1	-	1	0	1	-	-		
-	-	1	1	1	1	0	0	1	-	1	-	1	-	1	0	1	0	0	0	-	0	1	1	0	-	-	1	0	1	1	0	1	-	-		
0	-	0	1	1	1	-	1	-	1	1	0	0	0	0	1	-	-	1	1	0	0	1	-	1	1	0	1	-	-	-	1	0	1	-		
1	0	-	0	1	1	1	-	1	-	0	1	0	0	0	1	1	-	-	1	1	0	0	1	-	-	1	0	1	-	-	1	0	1	-		
1	1	0	-	0	-	1	1	-	1	0	0	1	0	0	1	1	1	-	-	-	1	0	0	1	-	-	1	0	1	-	1	0	1	-		
0	1	1	0	-	1	-	1	1	-	0	0	0	1	0	-	1	1	1	-	1	-	1	0	0	1	-	-	1	0	-	1	0	1	-		
-	0	1	1	0	-	1	-	1	1	0	0	0	0	1	-	-	1	1	1	0	1	-	1	0	0	1	-	-	1	-	1	0	1	-		
0	1	-	1	0	0	0	0	1	0	1	1	1	-	-	-	0	1	1	0	-	1	1	-	1	-	1	0	1	-	-	-	1	0	1		
0	0	1	-	1	0	0	0	0	1	-	1	1	1	-	0	-	0	1	1	1	-	1	1	-	-	-	1	0	1	-	-	1	0	1		
1	0	0	1	-	1	0	0	0	0	-	-	1	1	1	1	0	-	0	1	-	1	-	1	-	1	1	1	-	-	1	0	-	-	1	0	1
-	1	0	0	1	0	1	0	0	0	1	-	-	1	1	1	1	0	-	0	1	-	1	-	1	0	1	-	-	1	-	-	1	0	1		
1	-	1	0	0	0	0	1	0	0	1	1	-	-	1	0	1	1	0	-	1	1	-	1	-	1	0	1	-	-	-	-					

A symmetric $W(35, 25)$, with $-$ denoting -1 .

B Choosing $\Lambda = 0$

0	-	1	1	-	0	1	1	-	-	0	0	0	1	1	-	0	1	-	1	0	1	-	-	0	0	-	0	1	1	1	1	1		
-	0	-	1	1	-	0	1	1	-	1	-	0	0	0	-	1	-	0	1	-	1	0	1	-	1	0	0	-	0	1	1	1	1	
1	-	0	-	1	-	-	0	1	1	0	1	-	0	0	1	-	1	-	0	-	-	1	0	1	0	1	0	0	-	1	1	1	1	
1	1	-	0	-	1	-	-	0	1	0	0	1	-	0	0	1	-	1	-	1	-	-	1	0	-	0	1	0	0	1	1	1	1	
-	1	1	-	0	1	1	-	-	0	0	0	0	1	-	-	0	1	-	1	0	1	-	-	1	0	-	0	1	0	1	1	1	1	
0	-	-	1	1	1	0	0	0	0	1	1	-	1	-	-	1	1	1	-	1	-	1	0	0	1	-	0	1	0	0	1	-	-	1
1	0	-	-	1	0	1	0	0	0	-	1	1	-	1	-	-	1	1	1	0	1	-	1	0	0	1	-	0	1	0	1	-	-	1
1	1	0	-	-	0	0	1	0	0	1	-	1	1	-	1	-	-	1	1	0	0	1	-	1	1	0	1	-	0	0	1	-	-	1
-	1	1	0	-	0	0	0	1	0	-	1	-	1	1	1	1	-	-	1	1	0	0	1	-	0	1	0	1	-	0	1	-	-	1
-	-	1	1	0	0	0	0	0	1	1	-	1	-	1	1	1	1	-	-	-	1	0	0	1	-	0	1	0	1	0	1	-	-	1
-	1	0	0	0	1	-	1	-	1	1	1	-	-	1	0	0	-	1	1	-	0	1	0	1	0	0	-	1	1	1	0	1	-	-
0	-	1	0	0	1	1	-	1	-	1	1	1	-	-	1	0	0	-	1	1	-	0	1	0	1	0	0	-	1	1	0	1	-	-
0	0	-	1	0	-	1	1	-	1	-	1	1	1	-	1	1	0	0	-	0	1	-	0	1	1	1	0	0	-	1	0	1	-	-
0	0	0	-	1	1	-	1	1	-	-	-	1	1	1	-	1	1	0	0	1	0	1	-	0	-	1	1	0	0	1	0	1	-	-
1	0	0	0	-	-	1	-	1	1	1	-	-	1	1	0	-	1	1	0	0	1	0	1	-	0	-	1	1	0	1	0	1	-	-
1	-	1	0	-	-	-	1	1	1	0	1	1	-	0	-	1	0	0	1	1	1	0	-	0	1	-	0	1	0	-	1	0	1	-
-	1	-	1	0	1	-	-	1	1	0	0	1	1	-	1	-	1	0	0	0	1	1	0	-	0	1	-	0	1	-	1	0	1	-
0	-	1	-	1	1	1	-	-	1	-	0	0	1	1	0	1	-	1	0	-	0	1	1	0	1	0	1	-	0	-	1	0	1	-
1	0	-	1	-	1	1	1	-	-	1	-	0	0	1	0	0	1	-	1	0	-	0	1	1	0	1	0	1	-	1	0	1	-	-
-	1	0	-	1	-	1	1	1	-	1	1	-	0	0	1	0	0	1	-	1	0	-	0	1	-	0	1	0	1	-	1	0	1	-
1	-	-	1</																															

An alternative symmetric $W(35, 25)$ matrix where $\Lambda = \mathbf{0}$.

C The $\langle 41 \rangle$ representative

0	1	-	1	-	1	0	0	0	1	0	-	1	-	-	1	0	1	-	0	-	1	1	1	0	-	0	0	1	1	1	1	1		
1	0	1	-	0	-	1	0	0	-	1	0	-	1	-	-	1	0	1	1	0	-	-	1	0	1	0	-	0	1	1	1	1		
-	1	0	1	-	0	0	-	1	0	1	-	1	0	-	1	-	-	1	0	1	1	0	-	-	0	0	1	0	-	1	1	1		
-	-	1	0	1	0	0	0	-	1	-	1	-	1	0	0	1	-	-	1	-	1	1	0	-	-	0	0	1	0	1	1	1		
1	-	-	1	0	1	0	0	0	-	0	-	1	-	1	1	0	1	-	-	-	-	1	1	0	0	-	0	0	1	1	1	1		
-	0	0	0	1	1	-	1	1	-	1	-	-	1	1	0	1	1	-	0	1	0	1	-	0	1	1	0	0	-	0	1	-	-	
1	-	0	0	0	-	1	-	1	1	1	1	-	-	1	0	0	1	1	-	0	1	0	1	-	-	1	1	0	0	0	1	-	-	
0	1	-	0	0	1	-	1	-	1	1	1	1	-	-	-	0	0	1	1	-	0	1	0	1	0	-	1	1	0	0	1	-	-	
0	0	1	-	0	1	1	-	1	-	-	1	1	1	-	1	-	0	0	1	1	-	0	1	0	0	0	-	1	1	0	1	-	-	
0	0	0	1	-	-	1	1	-	1	-	-	1	1	1	1	1	-	0	0	0	1	-	0	1	1	0	0	-	1	0	1	-	-	
1	-	1	-	0	1	1	1	-	-	1	0	0	0	0	1	1	-	1	-	1	0	0	-	1	-	0	1	1	0	1	0	1	-	
0	1	-	1	-	-	1	1	1	-	0	1	0	0	0	-	1	1	-	1	1	1	0	0	-	0	-	0	1	1	1	0	1	-	
-	0	1	-	1	-	-	1	1	1	0	0	1	0	0	1	-	1	1	-	-	1	1	0	0	1	0	-	0	1	1	0	1	-	
1	-	0	1	-	1	-	-	1	1	0	0	0	1	0	-	1	-	1	1	0	-	1	1	0	1	1	0	-	0	1	0	1	-	
-	1	-	0	1	1	1	-	-	1	0	0	0	0	1	1	-	1	-	1	0	0	-	1	1	0	1	1	0	-	1	0	1	-	
-	-	1	0	1	0	0	-	1	1	1	-	1	-	1	-	1	0	0	1	1	0	-	0	1	0	-	0	1	1	-	1	0	1	-
1	-	-	1	0	1	0	0	-	1	1	1	-	1	-	1	-	1	0	0	1	1	0	-	0	1	0	-	0	1	-	1	0	1	-
0	1	-	-	1	1	1	0	0	-	-	1	1	-	1	0	1	-	1	0	0	1	1	0	-	1	1	0	-	0	-	1	0	1	-
1	0	1	-	-	-	1	1	0	0	1	-	1	1	-	0	0	1	-	1	-	0	1	1	0	0	1	1	0	-	-	1	0	1	-
-	1	0	1	-	0	-	1	1	0	-	1	-	1	1	1	0	0	1	-	0	-	0	1	1	-	0	1	1	0	-	1	0	1	-
0	1	1	-	-	1	0	-	1	0	1	1	-	0	0	1	1																		

A symmetric $W(35, 25)$ representative of $\langle 41 \rangle$.

E A neater $\Lambda = J$ representative

There is a better-structured $W_{\Lambda=J}$ representative with this block diagram:

$$\begin{pmatrix} \beta_1 & \beta_2 & \beta_3 & \text{id} & \beta_4 & \beta_5 & \psi_1 \\ \beta_2 & \beta_3 & \text{id} & \beta_4 & \beta_1 & \beta_5 & \psi_2 \\ \beta_3 & \text{id} & \beta_4 & \beta_1 & \beta_2 & \beta_5 & \psi_3 \\ \text{id} & \beta_4 & \beta_1 & \beta_2 & \beta_3 & \beta_5 & \psi_4 \\ \beta_4 & \beta_1 & \beta_2 & \beta_3 & \text{id} & \beta_5 & \psi_5 \\ \beta_5 & \beta_5 & \beta_5 & \beta_5 & \beta_5 & J & \mathbf{0} \\ \psi_1^\top & \psi_2^\top & \psi_3^\top & \psi_4^\top & \psi_5^\top & \mathbf{0} & J \end{pmatrix}$$

The following *Wolfram Language* code, run with Mathematica 15.0, generates this arrangement. A standalone copy is `scripts/neater_lambda_j.wls` in the companion repository.

```
ClearAll["Global`*"];
swizzled[v_] := Table[v[[Mod[i + j - 2, 5] + 1]], {i, 5}, {j, 5}];
border[core_, edge_, corner_] := ArrayFlatten[{
  {core, ConstantArray[edge, {5, 1}]},
  {ConstantArray[edge, {1, 5}], {{corner}}}}
];
a = {1, 1, -1, 1, -1};
b = {-1, 1, 1, 0, 0};
c = {1, -1, 0, 0, 1};
a0 = border[swizzled[a], 0, 1];
a1 = border[swizzled[b], 1, 1];
a2 = border[swizzled[c], -1, 1];
f0 = {{0, 1, -1, -1, 1, 0},
  {1, 0, 1, -1, -1, 0},
  {-1, 1, 0, 1, -1, 0},
  {-1, -1, 1, 0, 1, 0},
  {1, -1, -1, 1, 0, 0}};
f = f0[[All, {1, 2, 4, 5, 3, 6}]];
S = RotateRight[IdentityMatrix[5]];
moving = KroneckerProduct[a0, IdentityMatrix[5]] +
  KroneckerProduct[a1, S + Transpose[S]] +
  KroneckerProduct[a2, MatrixPower[S, 2] + Transpose[MatrixPower[S, 2]]];
rolling = KroneckerProduct[f, ConstantArray[1, {1, 5}]];
w35 = ArrayFlatten[{
  {moving, Transpose[rolling]},
  {rolling, ConstantArray[1, {5, 5}]}
];
On[Assert];
Assert[SymmetricMatrixQ[w35]];
Assert[Total[Abs@w35] == ConstantArray[25, 35]];
Assert[w35 . Transpose[w35] == 25 IdentityMatrix[35]];
MatrixPlot[w35, Mesh -> {6, 6}, MeshStyle -> Black]
```

F Code for finding a $W(35, 25)$ by CP-SAT

I include the Python 3 code that uses the OR-Tools 9.15.6755 CP-SAT solver [10] and NumPy 2.5.2 to find $W_{\Lambda=J}$ and $W_{\Lambda=0}$. A standalone copy is `scripts/find_w35_cpsat.py` in the companion repository. On a desktop workstation with an Intel® Core™ i9-10940X CPU clocked at 3.30 GHz and 160 GB of DDR4 RAM, this code finds a solution in under a minute with arguments

```
--lambda-block 1 --workers 16.
```

```
# Author: Christopher Munro 2026.
# This code is released under the Creative Commons Attribution (CC BY) 4.0 license.
# This lets others distribute, remix, adapt, and build upon this work, even commercially,
# as long as they credit the original creator.
# https://creativecommons.org/share-your-work/use-remix/cc-licenses/

import argparse
import numpy as np
from ortools.sat.python import cp_model

def check(condition, message):
    if not condition:
        raise RuntimeError(message)

def blocks(lambda_block):
    if lambda_block == 0:
        C = np.array(
            [
                [1, 0, 1, -1, -1, 1],
                [1, 1, 0, 1, -1, -1],
                [1, -1, 1, 0, 1, -1],
                [1, -1, -1, 1, 0, 1],
                [1, 1, -1, -1, 1, 0],
            ],
            dtype=np.int64,
        )
        E_block_sums = np.zeros((6, 6), dtype=np.int64)
        E_block_sums[1:, 1:] = 1
        A = np.zeros((5, 5), dtype=np.int64)
    else:
        C = np.array(
            [
                [0, 1, -1, -1, 1, 0],
                [1, 0, 1, -1, -1, 0],
                [-1, 1, 0, 1, -1, 0],
                [-1, -1, 1, 0, 1, 0],
                [1, -1, -1, 1, 0, 0],
            ],
            dtype=np.int64,
        )
        E_block_sums = np.zeros((6, 6), dtype=np.int64)
        E_block_sums[:5, :5] = 1
        E_block_sums[5, 5] = 5
        A = np.ones((5, 5), dtype=np.int64)

    #  $5CC^T + AA^T = 25 I_5$ 
    check(
        np.array_equal(
            5 * C @ C.T + A @ A.T,
            25 * np.eye(5, dtype=np.int64),
        ),
        "C and A do not satisfy the fixed-row equation",
    )
    return C, E_block_sums, A

def solve(lambda_block, workers, seed):
    C, E_block_sums, A = blocks(lambda_block)

    # Step 1: repeat each column of C five times to form  $\Psi$ .
    Psi = np.repeat(C.T, 5, axis=0)
    C_column_products = C.T @ C
```



```

model = cp_model.CpModel()

# Step 2: s_ij is the block seed for Cyc(s_ij) in  $\Sigma$ .
block_seeds = {}
for block_row in range(6):
    for block_column in range(block_row, 6):
        block_seeds[block_row, block_column] = tuple(
            model.new_int_var(
                -1,
                1,
                f"s_{block_row}_{block_column}_{seed_position}",
            )
            for seed_position in range(5)
        )

def seed_entry(block_row, block_column, seed_position):
    seed_position %= 5
    if block_row <= block_column:
        return block_seeds[block_row, block_column][seed_position]
    return block_seeds[block_column, block_row][(-seed_position) % 5]

# Symmetry gives s_ji by reversing s_ij
for block in range(6):
    model.add(seed_entry(block, block, 1) == seed_entry(block, block, 4))
    model.add(seed_entry(block, block, 2) == seed_entry(block, block, 3))

# Every row of Cyc(s_ij) has sum sum(s_ij).
for block_row in range(6):
    for block_column in range(block_row, 6):
        model.add(
            sum(
                seed_entry(block_row, block_column, seed_position)
                for seed_position in range(5)
            )
            == int( $\Sigma$ _block_sums[block_row, block_column])
        )

# This is the circulant form of  $\Sigma \Sigma^T + \Psi \Psi^T = 25 I_{30}$ .
# row_displacement is the second row's position minus the first row's
# position within their respective 5x5 block rows.
for first_row_block in range(6):
    for second_row_block in range(first_row_block, 6):
        for row_displacement in range(5):
            products = []
            for column_block in range(6):
                for seed_position in range(5):
                    product = model.new_int_var(
                        -1,
                        1,
                        "p_"
                        f"{first_row_block}_{second_row_block}_"
                        f"{row_displacement}_{column_block}_{seed_position}",
                    )
                    model.add_multiplication_equality(
                        product,
                        [
                            seed_entry(
                                first_row_block,
                                column_block,
                                seed_position,
                            ),
                            seed_entry(
                                second_row_block,
                                column_block,
                                seed_position - row_displacement,
                            ),
                        ],
                    )
                    products.append(product)

required_inner_product = (
    25
    if first_row_block == second_row_block
    and row_displacement == 0
    else 0

```

```

    )
    model.add(
        sum(products)
        + int(C_column_products[first_row_block, second_row_block])
        == required_inner_product
    )

solver = cp_model.CpSolver()
solver.parameters.num_search_workers = workers
solver.parameters.random_seed = seed

status = solver.solve(model)
if status not in (cp_model.OPTIMAL, cp_model.FEASIBLE):
    raise SystemExit(
        f"no W(35,25) found: solver status {solver.status_name(status)}"
    )

# Step 2, continued: build the circulant blocks from the solved seeds.
Σ = np.zeros((30, 30), dtype=np.int64)
for block_row in range(6):
    for block_column in range(6):
        for row_in_block in range(5):
            for column_in_block in range(5):
                seed_position = (column_in_block - row_in_block) % 5
                Σ[
                    5 * block_row + row_in_block,
                    5 * block_column + column_in_block,
                ] = solver.value(
                    seed_entry(block_row, block_column, seed_position)
                )

# Step 3: verify the two block equations for  $W W^T = 25 I_{35}$ .
target_30 = 25 * np.eye(30, dtype=np.int64)
check(
    np.array_equal(Σ @ Σ.T + Ψ @ Ψ.T, target_30),
    "ΣΣ.T + ΨΨ.T is not 25 I_30",
)
check(
    np.array_equal(Σ @ Ψ + Ψ @ Λ, np.zeros((30, 5), dtype=np.int64)),
    "ΣΨ + ΨΛ is not zero",
)

# Assemble the block matrix:  $W = ((Σ, Ψ), (Ψ^T, Λ))$ .
W = np.block([[Σ, Ψ], [Ψ.T, Λ]])
target_35 = 25 * np.eye(35, dtype=np.int64)
check(np.array_equal(W, W.T), "W is not symmetric")
check(np.all(np.isin(W, (-1, 0, 1))), "W has a non-ternary entry")
check(np.array_equal(W @ W.T, target_35), "W W^T is not 25 I_35")
check(np.array_equal(W.T @ W, target_35), "W^T W is not 25 I_35")
output_path = f"w35-{lambda_block}.txt"
np.savetxt(output_path, W, fmt="%d")
print(f"done: {output_path}", flush=True)

if __name__ == "__main__":
    parser = argparse.ArgumentParser()
    parser.add_argument("--lambda-block", type=int, choices=(0, 1), required=True)
    parser.add_argument("--workers", type=int, default=16)
    parser.add_argument("--seed", type=int, default=1)
    arguments = parser.parse_args()

    solve(
        arguments.lambda_block,
        arguments.workers,
        arguments.seed,
    )

```

G Code for signed-equivalence checking

I used this *Wolfram Language* code to check for signed-permutation equivalence. A standalone copy is `scripts/wm_equivalent.wls` in the companion repository.

```
ClearAll["Global`*"];

g[w_, t_, signed_] :=
  With[{n = Length@w},
    Graph@Join[
      Flatten@Table[
        pr[t, i] \[DirectedEdge] r[t, i, s], {i, n}, {s, {-1, 1}},
      Flatten@Table[
        c[t, j, s] \[DirectedEdge] pc[t, j], {j, n}, {s, {-1, 1}},
      Flatten@Table[
        If[w[[i, j]] == 0, Nothing,
          r[t, i, s] \[DirectedEdge] c[t, j, s w[[i, j]]], {i, n}, {j,
            n}, {s, {-1, 1}},
        If[signed, {},
          Join[Table[rm[t, i] \[DirectedEdge] r[t, i, 1], {i, n}],
            Table[c[t, j, 1] \[DirectedEdge] cm[t, j], {j, n}]]]]]]
WMEquivalent[a_, b_, signed_ : True] :=
  Module[{n = Length@a, f, ri, ci, p, q},
    f = FindGraphIsomorphism[g[a, 0, signed], g[b, 1, signed], 1];
    If[f === {}, Return["DISTINCT"]];
    f = First@f;
    ri = Table[f[r[0, i, 1]] /. r[1, j_, s_] :> {j, s}, {i, n}];
    ci = Table[f[c[0, i, 1]] /. c[1, j_, s_] :> {j, s}, {i, n}];
    p = SparseArray[Table[{ri[[i, 1]], i} -> ri[[i, 2]], {i, n}], {n, n}];
    q = SparseArray[Table[{ci[[i, 1]], i} -> ci[[i, 2]], {i, n}], {n, n}];
    {p, q}]
```

An example of usage:

$\text{WMEquivalent}(W_{\langle 41 \rangle}, W_{\langle 38 \rangle}) = \text{DISTINCT}$ and $\text{WMEquivalent}(W_{\langle 40 \rangle}, W_{\langle 31 \rangle}) = \text{DISTINCT}$.

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